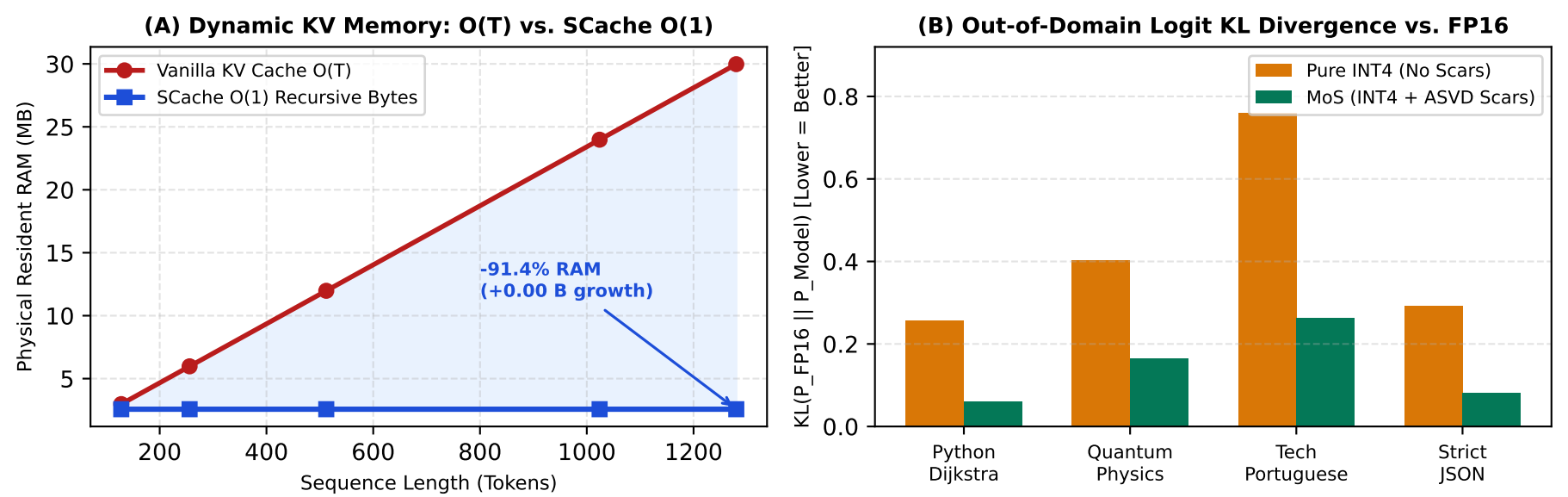
(A) Dynamic KV Memory: O(T) vs. SCache O(1)
Vanilla KV Cache O(T)
SCache O(1) Recursive Bytes
-91.4% RAM
(+0.00 B growth)
Physical Resident RAM (MB)
Sequence Length (Tokens)
30
25
20
15
10
5
200
400
600
800
1000
1200
(B) Out-of-Domain Logit KL Divergence vs. FP16
Pure INT4 (No Scars)
MoS (INT4 + ASVD Scars)
KL(P_FP16 || P_Model) [Lower = Better]
0.8
0.6
0.4
0.2
0.0
Python
Dijkstra
Quantum
Physics
Tech
Portuguese
Strict
JSON